

Break Your Own Site — Hardening & QA Audit Report

Rigorous Edge-Case Stress Testing, Form Hardening, SEO Metadata Ingestion & Performance Auditing

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Track: General AI Fluency (Week 09)

Production Portfolio: [abdulsamiuthwal-portfolio.vercel.app](#)

GitHub Repository: [abdulsamiuthwal-eng/Flyrank-assignments](#)

1. Executive Summary & Audit Scope

Rather than relying on happy-path demonstrations, this audit systematically tested the application against empty inputs, garbage text, rapid multi-click submissions, missing social preview tags, and cross-device viewports. All identified high-priority vulnerabilities have been remediated in code.

2. "Where It Breaks" Stress Testing Matrix

Scenario	Trigger Action	Pre-Hardening Behavior	Triage	Fix / Remediation Applied
1. Empty Form	Clicked Submit with empty inputs	Browser default native alert	Fix-Now	Added input trimming, regex validation, and inline status feedback element.
2. Double Click	Clicked Submit twice in <300ms	Sent duplicate HTTP requests	Fix-Now	Added submit locking (isSubmitting flag) & button disabling.
3. Invalid Email	Entered test@ or invalid format	Allowed submission / raw alert	Fix-Now	RFC-compliant email regex validation with user feedback.
4. Space Input	Filled name with spaces " "	Accepted spaces as valid string	Fix-Now	Enforced .trim() sanitization across all input fields.
5. Social Meta	Link shared on WhatsApp / LinkedIn	Blank preview card	Fix-Now	Injected OpenGraph (og:title, og:image), Twitter Cards & Schema JSON-LD.
6. API Throttling	>3 consecutive API calls / sec	Temporary HTTP 429 throttling	Limitation	Documented rate limit behavior with exponential backoff notice.

3. Performance & Speed Audit Results

Audit Category	Metric / Tool	Result Score	Status
Page Speed Score	Lighthouse / PageSpeed	98 / 100	✅ PASS
First Contentful Paint	Navigation Timing API	0.4 seconds	✅ PASS
Largest Contentful Paint	Performance Observer	0.8 seconds	✅ PASS
Agent Response Latency	Flask REST API / arXiv	681ms – 898ms	✅ PASS

4. Injected SEO & Social Preview Metadata

```
<title>Abdul Sami Uthwal | AI Systems & Automation Engineering Portfolio</title>
<meta name="description" content="Official FlyRank portfolio of Abdul Sami Uthwal –
AI Systems & Automation Engineer...">
<meta property="og:type" content="website">
<meta property="og:url" content="https://abdulsamiuthwal-portfolio.vercel.app/">
<meta property="og:title" content="Abdul Sami Uthwal – AI Systems & Automation
Engineer">
<meta property="og:image" content="https://abdulsamiuthwal-
portfolio.vercel.app/sitemap_sketch.png">
<meta name="twitter:card" content="summary_large_image">
```